YOON, KUK-JIN

Professor

Visual Intelligence Laboratory
Department of Mechanical Engineering
Korea Advanced Institute of Science and Technology (KAIST)
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Education

Ph.D. in Electrical Engineering and Computer Science, KAIST, Korea — Feb. 2006
Thesis: Robust Correspondence Search under Photometric Variations and Image Ambiguity

M.S. in Electrical Engineering and Computer Science, KAIST, Korea — Feb. 2000
Thesis: Moving Object Segmentation with an Accurate Boundary Using Color and Motion in Color Image Sequence

B.S. in Electrical Engineering and Computer Science, KAIST, Korea — Feb. 1998

Professional Experience

Korea Advanced Institute of Science and Technology (KAIST) — Feb. 2018 – Present
Professor, Department of Mechanical Engineering

- Dean of College of AI (Jul. 2026 – Present)
- Director of KAIST AI Institutes (Jul. 2026 – Present)
- Interim Dean of College of AI (Dec. 2025 – Jul. 2026)
- Deputy Director of KAIST AI Institutes (Feb. 2025 – Jul. 2026)
- Head of KAIST Robotics Program (Feb. 2023 – Jul. 2023)
- Affiliated: Kim Jaechul Graduate School of AI, KAIST Institute for Robotics, Division of Future Vehicle

Samsung Research America, Mountain View, CA, US — Aug. 2023 – Jul. 2024
Visiting Scholar

Korea Institute of Science and Technology (KIST) — Sep. 2013 – Aug. 2014
Visiting Scholar

Gwangju Institute of Science and Technology (GIST) — Aug. 2008 – Feb. 2018
Assistant & Associate Professor, School of Electrical Engineering and Computer Science

- Secretary of GIST Faculty Assembly (2016 – 2017)

INRIA Rhône-Alpes, France — Aug. 2006 – Aug. 2008
Post-doctoral Fellow (Perception Team)

KAIST, Robotics and Computer Vision Lab. — Mar. 2006 – May 2006
Post-doctoral Researcher

Corporate & Technical Advisories

External Advisory Committee, Samsung Electronics — Mar. 2022 – Dec. 2022

Technical Adviser, 42dot	Feb. 2022 – Aug. 2022
Technical Adviser, Avikus	Feb. 2021 – Aug. 2021
Technical Adviser, Samsung Electronics (Visual Display Division)	Apr. 2017 – Dec. 2017
Technical Adviser, NAVER Labs (Mobility Team)	Mar. 2017 – Aug. 2017

Research Interests

Computer vision, machine learning, pattern recognition

- **Visual intelligence** for autonomous systems and robotics
- **Event-based and multimodal visual sensing** (event, 360°, RGB-D, LiDAR, radar)
- **World and foundation models** for embodied AI and autonomous driving
- **Geometry-aware 3D scene understanding** (multi-view reconstruction, SLAM, VO)
- **Multi-object detection and tracking**, and motion understanding
- **Multi-sensor fusion and learning** under weak or limited supervision

Honors & Awards

- **KAIST Education Innovation Award**, 2026.
- **KAIST Major Achievements in 14 Future Technology Areas** (Autonomous Driving), 2026.
- **KAIST Impact Research Award**, 2025.
- **IEEE/CVF WACV Test of Time Award**, 2025.
- **Bronze Prize (as an Advisor)**: Samsung HumanTech Paper Award, 2023.
- **Best Paper Award (Gold Prize)**: 35th Workshop on Image Processing and Image Understanding, 2023.
- **Best Paper Awards (Grand and Gold Prizes)**: 34th Workshop on Image Processing and Image Understanding, 2022.
- **Technology Innovation Award, College of Engineering, KAIST**, 2022
- **Selection of KAIST's Top 10 Research Achievements**, 2022.
- **KI (KAIST Institute) Convergence Researcher Award**, 2021.
- **The 1st and 3rd Place** at the Event-based Stereo Challenge in CVPRW 2021, 2021.
- **Commendation from the Korea Minister of Science and ICT** in Recognition of Contributions in the field of Artificial Intelligence (인공지능산업발전유공 과학기술정보통신부장관 표창), Dec. 2020.
- **Sang-Uk Lee Prize (test-of-time award)** at Korean Conference on Computer Vision by Korean Computer Vision Society, 2020.
- **Best Paper Award**: Korea Software Congress 2019 by The Korean Institute of Information Scientists and Engineers, 2019.
- **Best Paper Awards (Grand and Bronze Prizes)**: 31th Workshop on Image Processing and Image Understanding, 2019.
- **Best Student Paper Award (as an Advisor)**: IW-FCV 2018, 2018.
- **Best Paper Award (Silver Prize) and Best Poster Paper Award**: 30th Workshop on Image Processing and Image Understanding, 2018.
- **Silver Prize (as an Advisor)**: Samsung HumanTech Paper Award, 2017.
- **Best Poster Presentation Award (as an Advisor)**: IW-FCV 2017, 2017.
- **Outstanding Reviewer**, ECCV 2016, 2016.

- **Best Paper Award**: 28th Workshop on Image Processing and Image Understanding, 2016.
- **Bronze Prize (as an Advisor)**: Samsung HumanTech Paper Award, 2016.
- **Silver Prize (as an Advisor)**: Samsung HumanTech Paper Award, 2015.
- **Participation Prize (as an Advisor)**: Samsung HumanTech Paper Award, 2015.
- **The 1st Place** at the 1st Multi-object Tracking Challenge (MOT Competition sponsored by Daimler), 2015.
- **Best Paper Award**: 9th Korea Robotics Society Annual Conference, 2014.
- **Best Paper Award**: 26th Signal Processing Conference by The Institute of Electronics and Information Engineers, 2014.
- **Best Paper Award**: 26th Workshop on Image Processing and Image Understanding, 2014.
- **Silver Prize (as an Advisor)**: Samsung HumanTech Paper Award, 2014.
- **Bronze Prize (as an Advisor)**: Samsung HumanTech Paper Award, 2014.
- **Silver Prize (as an Advisor)**: Samsung HumanTech Paper Award, 2012.
- **Grants to Post-Doctoral Fellows** by INRIA, 2006.
- **Government Grant to Post-Doctoral Fellows** by Korea Research Foundation, 2006.
- **Silver Prize**: Samsung HumanTech Paper Award, 2006.
- **Top 10% among the Accepted Papers**: ICIP, 2005.
- **Bronze Prize**: Samsung HumanTech Paper Award, 2005.
- **Research Prize**: The Fifth Korean Intelligent Robot Contest, 2003.
- **The 3rd Place**: Best Poster Award in Photonics Boston, 2001.

Professional and Public Service

Member, Special Committee for a Talent-Power Nation, Presidential Council on National Education Commission	Nov. 2025 – May 2026
Member, Presidential Council on National Artificial Intelligence Strategy	Sep. 2025 – Present
Member, Committee on Autonomous Vehicle Accident Investigation, KATRI/KOTSA	Jul. 2025 – Jul. 2027
Advisory Committee Member, 2nd Operations Command, Republic of Korea Army	Mar. 2021 – Dec. 2025
Steering Committee Member, National Strategic Projects on VR/AR	Mar. 2017 – Dec. 2018
Steering Committee Member, Korea Culture Technology Institute	Dec. 2015 – Feb. 2018

Academic Service

International

- **Associate Editor:** IEEE TPAMI (since Apr. 2023), CVIU (since Mar. 2023), IJAT (since 2017)
- **Area Chair / Senior AC:** ICLR 2027, AAAI 2027, WACV 2027, NeurIPS 2026, ECCV 2026, ICML 2026, CVPR 2026, AAAI 2026, WACV 2026, NeurIPS 2025, ICCV 2025 (Lead AC), ICML 2025, NeurIPS 2024, ECCV 2024, WACV 2024, ICCV 2023, CVPR 2022, WACV 2022, ICCV 2021, CVPR 2020, ICCV 2019, ACM MM 2019, etc.
- **Organizing Committee Member:** IEEE Intelligent Vehicles Symposium (IV 2024), ICCV 2019
- **Program Co-chair:** ICCV Workshop and Challenge on Comprehensive Video Understanding in the Wild (CoView 2019)
- **General Co-chair:** International Symposium on Future Mobility (ISFM) 2019

Domestic

- **Organizer / Chair:** HMG Autonomous Driving Challenge 2025 (Organizer), KCCV 2022 (Program Chair)
- **Board Member / Officer:** Korean Computer Vision Society (KCVS) (since 2016), The Korea Robotics Society (2016)
- **Program Committee:** KCCV (since 2014), IPIU (2010 – 2018)
- **Editor:** The Journal of Korea Robotics Society (2011 – 2014), The Journal of Korea Information Processing Society (2009 – 2014)

Publications (International)

*Note: Underline indicates first or corresponding author and * indicates equal contribution.*

SCI Journals

74. Jinho Kim and **Kuk-Jin Yoon**, "Encoder-Level Temporal Fusion Transformers for Robust Multi-Object Tracking," IEEE Access, 2026.
73. Wooseong Jeong, Jegyeong Cho, Youngho Yoon, Jaeyoung Lee, **Kuk-Jin Yoon**, "Stabilizing Multi-Task Latent Spaces: Recursive Refinement With Coordinators in Partially Labeled Learning," vol. 14, IEEE Access, 2026.
72. Mingu Kang, Joon-Kyu Han, Kichul Lee, Jaeseok Jeong, Chanyoung Yoo, Jeong Woo Jeon, Byongwoo Park, Wonho Choi, Junseong Ahn, **Kuk-Jin Yoon**, Cheol Seong Hwang, and Inkyu Park, "Neuromorphic Olfaction with Ultralow-power Gas Sensors and Ovonic Threshold Switch", Science Advances, vol. 11, no. 39, 2025.
71. Hyeonseong Kim*, Yoonsu Kang*, Changgyoon Oh, and **Kuk-Jin Yoon**, "Towards Enhanced Representation Learning for Single-Source Domain Generalization in LiDAR Semantic Segmentation," IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2025.
70. Donghyeong Lee, Seokwan Roh, Jaewoo Jeong, **Kuk-Jin Yoon**, Jungchul Lee, and Youngsuk Nam, "Analysis of Multiscale Condensation Phenomena Using a Zero-Shot Computer Vision Framework," Advanced Science, 2025.
69. Jae-Young Kang*, Hoonhee Cho*, and **Kuk-Jin Yoon**, "Temporal Stereo Matching from Event Cameras via Joint Learning with Stereoscopic Flow," IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2025.
68. Inkyu Shin, Qihang Yu, Ming-Hsuan Yang, Xiaohui Shen, In So Kweon, **Kuk-Jin Yoon**, and Liang-Chieh Chen, "Enhancing Temporal Consistency in Video Editing by Reconstructing Videos with 3D Gaussian Splatting," Transactions on Machine Learning Research (TMLR) 2025
67. Byeongju Lee, Mingu Kang, Kichul Lee, Yujeong Chae, **Kuk-Jin Yoon**, Dae-Sik Lee, and Inkyu Park, "Multigas Identification by Temperature-Modulated Operation of a Single Anodic Aluminum Oxide Gas Sensor Platform and Deep Learning Algorithm," ACS Sensors, vol. 10, no. 2, pp. 954-964, 2025.
66. Wooseong Jeong, Jihun Kim, Hyeokjun Kweon, and **Kuk-Jin Yoon**, "Multi-View 3D Scene Abstraction From Drone-Captured RGB Images," IEEE Access, vol. 13, pp. 27641-27656, 2025.
65. Minjie Liu, Hongjian Wang, **Kuk-Jin Yoon**, Lin Wang, "Disentangled Cross-modal Fusion for Event-guided Image Super-Resolution," IEEE Transactions on Artificial Intelligence, accepted, 2024.
64. Seongyong Ahn, Inwook Shim, Jihong Min, **Kuk-Jin Yoon**, "EasyFuse: Easy-to-Learn Visible and Infrared Image Fusion Framework based on Unpaired Set," Pattern Recognition Letters, 2023.
63. Incheol Cho, Kichul Lee, Young Chul Sim, Jaeseok Jeong, Minkyu Cho, Heechan Jung, Mingu Kang, Yong-Hoon Cho, Seung Chul Ha, and **Kuk-Jin Yoon***, Inkyu Park* (*: co-corresponding), "Deep Learning-based Gas Identification by Time-variant Illumination of a Single Micro LED-embedded Gas Sensor," accepted to Light: Science & Applications, 2023.

62. Kichul Lee, Incheol Cho, Mingu Kang, Jaeseok Jeong, Minho Choi, Kie Young Woo, **Kuk-Jin Yoon***, Yong-Hoon Cho*, and Inkyu Park* (*: co-corresponding), "Ultra-Low-Power E-Nose System Based on Multi-Micro-LED-Integrated, Nanostructured Gas Sensors and Deep Learning," ACS Nano, 2023, vol. 17, no. 1, pp. 539–551.

61. Lin Wang and **Kuk-Jin Yoon**, "Deep Learning for HDR Imaging: State-of-the-Art and Future Trends," IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2022.

60. Lin Wang, Tae-Kyun Kim, and **Kuk-Jin Yoon**, "Joint Framework for Single Image Reconstruction and Super-Resolution With an Event Camera," IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2022.

59. Joon-Kyu Han, Mingu Kang, Jaeseok Jeong, Incheol Cho, Ji-Man Yu, **Kuk-Jin Yoon**, Inkyu Park*, Yang-Kyu Choi* (*: co-corresponding), "Artificial Olfactory Neuron for an In-Sensor Neuromorphic Nose," Advanced Science, vol. 9, no. 18, 2022.

58. Mingu Kang, Incheol Cho, Jaeho Park, Jaeseok Jeong, Kichul Lee, Byeongju Lee, Dionisio Del Orbe Henriquez, **Kuk-Jin Yoon***, Inkyu Park* (*: co-corresponding), "High Accuracy Real-Time Multi-Gas Identification by a Batch-Uniform Gas Sensor Array and Deep Learning Algorithm," ACS Sensors, vol. 7, no. 2, pp. 430-440, 2022.

57. Lin Wang and **Kuk-Jin Yoon**, "Semi-supervised Student-Teacher Learning for Single Image Super-Resolution," Pattern Recognition (PR), 2021.

56. S. Mohammad Mostafavi I., Yeongwoo Nam, Jonghyun Choi, and **Kuk-Jin Yoon**, "E2SRI: Learning to Super-Resolve Intensity Images From Events," IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2022.

55. Kwonyoung Ryu, Kang-il Lee, Jegyeong Cho, and **Kuk-Jin Yoon**, 'Scanline Resolution-invariant Depth Completion using a Single Image and Sparse LiDAR Point Cloud," IEEE Robotics and Automation Letters (RA-L), 2021.

54. Hoonhee Cho, Jaeseok Jeong, and **Kuk-Jin Yoon**, "EOMVS : Event-based Omnidirectional Multi-View Stereo," IEEE Robotics and Automation Letters (RA-L), 2021.

53. Ji-il Park, Yeongseok Lee, Eungyo Suh, Hyunyong Jeon, **Kuk-Jin Yoon***, and Kyung-Soo Kim*, "Improvement of Optical Flow Estimation by Using the Hampel Filter for Low-End Embedded Systems," IEEE Robotics and Automation Letters (RA-L), 2021.

52. Lin Wang and **Kuk-Jin Yoon**, "PSAT-GAN: Efficient Adversarial Attacks against Holistic Scene Understanding," IEEE Transactions on Image Processing (TIP), 2021.

51. Lin Wang and **Kuk-Jin Yoon**, "Knowledge Distillation and Student-TeacherLearning for Visual Intelligence: A Review and New Outlooks," IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2021.

50. S. Mohammad Mostafavi I., Lin Wang, and **Kuk-Jin Yoon**, "Learning to Reconstruct HDR Images from Events, with Applications to Depth and Flow," International Journal of Computer Vision (IJCV), 2021.

49. Taewoo Kim, Kyeongseob Song, Kwonyoung Ryu, and **Kuk-Jin Yoon**, "Loop-Net: Joint Unsupervised Disparity and Optical Flow Estimation of Stereo Videos with Spatiotemporal Loop Consistency," IEEE Robotics and Automation Letters (RA-L), 2020.

48. Yeon Kun Lee, Jaeseok Jeong*, Jong Seob Yun*, Won June Cho*, and **Kuk-Jin Yoon** (*: equal contribution), "SpherePHD: Applying CNNs on 360° Images with Non-Euclidean Spherical PolyHeDron Representation," IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2020.

47. Lin Wang, Wonjune Cho, and **Kuk-Jin Yoon**, "Deceiving Image-to-Image Translation Networks for Autonomous Driving with Adversarial Perturbations," IEEE Robotics and Automation Letters (RA-L), 2020.

46. Jeong-Kyun and **Kuk-Jin Yoon**, "Joint Estimation of Camera Orientation and Vanishing Points from Lines," International Journal of Computer Vision (IJCV), 2019.

45. Yeong-Jun Cho and **Kuk-Jin Yoon**, "Distance-based Camera Network Topology Inference for Person Re-identification," Pattern Recognition Letters, 2019.

44. Chang-Ryeol Lee and **Kuk-Jin Yoon**, “Confidence Analysis of Feature Points for Visual-Inertial Odometry of Urban Vehicles,” IET Intelligent Transport Systems, 2019.

43. Min-Gyu Park and **Kuk-Jin Yoon**, “As-Planar-As-Possible Depth Map Estimation," Computer Vision and Image Understanding (CVIU), 2019.

42. Yeong-Jun Cho, Su-A Kim, Jae-Han Park, Kyuewang Lee, and **Kuk-Jin Yoon**, “Joint Person Re-identification and Camera Network Topology Inference in Multiple Cameras,” Computer Vision and Image Understanding (CVIU), 2019.

41. Hanmu Park and **Kuk-Jin Yoon**, “Exploiting Multi-layer Graph Factorization for Multi-attributed Graph Matching,” Pattern Recognition Letters, 2019.

40. Hanmu Park and **Kuk-Jin Yoon**, “Consistent Multiple Graph Matching with Multi-layer Random Walks Synchronization,” Pattern Recognition Letters, 2019.

39. Min-Gyu Park and **Kuk-Jin Yoon**, “Learning and Selecting Confidence Measures for Robust Stereo Matching,", IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), vol. 41, no. 6, pp. 1397-1411, 2019.

38. Ju Hong Yoon, Chang-Ryeol Lee, Ming-Hsuan Yang, and **Kuk-Jin Yoon**, “Structural Constraint Data Association for Online Multi-Object Tracking,” International Journal of Computer Vision (IJCV), vol. 127, no. 1, pp. 1-21, 2019.

37. Chang-Ryeol Lee, Ju Hong Yoon, and **Kuk-Jin Yoon**, “Calibration and Noise Identification of a Rolling Shutter Camera and a Low-cost Inertial Measurement Unit,” Sensors, vol. 18, no. 7, 2018.

36. Yeong-Jun Cho and **Kuk-Jin Yoon**, “PAMM: Person Re-identification via Pose-aware Multi-shot Matching,” IEEE Transactions on Image Processing (TIP), vol. 27, no. 8, pp. 3739-3752, 2018.

35. Jeong-Kyun Lee and **Kuk-Jin Yoon**, “Temporally Consistent Road Surface Profile Estimation Using Stereo Vision," IEEE Transactions on Intelligent Transportation System (T-ITS), vol. 19, n o. 5, pp. 1618-1628, 2018.

34. Han-Mu Park and **Kuk-Jin Yoon**, “Multi-attributed Graph Matching with Multi-layer Graph Structure and Multi-layer Random Walks,” IEEE Transactions on Image Processing (TIP), vol. 27, no. 5, pp. 2314-2325, 2018.

33. Seung Hwan Bae and **Kuk-Jin Yoon**, “Confidence-Based Data Association and Discriminative Deep Appearance Learning for Robust Online Multi-Object Tracking,” IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), vol. 40, no. 3, pp. 595-610, 2018.

32. Han-Mu Park, Dae-Yong Cho, and **Kuk-Jin Yoon**, “Greedy Refinement of Object Proposals via Boundary-aligned Minimum Bounding Box Search," IET Computer Vision (CVI), vol. 12, no. 3, pp. 357-363, 2018.

31. Han-Mu Park, Se-Hoon Park, and **Kuk-Jin Yoon**, “Multi-object Tracking via Tracklet Confidence-Aided Relative Motion Analysis,” SPIE Journal of Electronic Imaging, 2017.

30. Hohyun Cho, Min-Koo Kang, Sangtae Ahn, Moonyoung Kwon, **Kuk-Jin Yoon**, Kiwoong Kim, and Sung Chan Jun, “Cognitive Response and Cortical Oscillatory Processing for Various Stereoscopic Depths - Simultaneous EEG/MEG Study," Journal of Integrative Neuroscience, 2017.

29. Min-Koo Kang, Hohyun Cho, Han- Mu Park, Sung Chan Jun, and **Kuk-Jin Yoon**, “A Wellness Platform for Stereoscopic 3D Video Systems Using EEG-based Visual Discomfort Evaluation Technology,” Applied Ergonomics, vol. 62, pp. 158-167, 2017.

28. Yeong-Jun Cho, Seung Hwan Bae, and **Kuk-Jin Yoon**, “Multi-Classier-based Automatic Polyp Detection in Endoscopic Images,” Journal of Medical and Biological Engineering, Published Online, Nov. 28, 2016.

27. Hohyun Cho, Min-Koo Kang, Sangtae Ahn, Moonyoung Kwon, **Kuk-Jin Yoon**, Kiwoong Kim, and Sung Chan Jun, “Cortical Responses and Shape Complexity of Stereoscopic Image – A Simultaneous EEG/MEG Study," NeuroSignals, vol. 24, no. 1, pp. 102–112, 2016.

26. Seung Hwan Bae, Jong-Youl Park, and **Kuk-Jin Yoon**, “Joint Estimation of Multi-Target SNR and Dynamic States in Cluttered Environment,” IET Radar, Sonar and Navigation, Published Online, Oct. 19, 2016.

25. Han-Mu Park and **Kuk-Jin Yoon**, “Encouraging Second-order Consistency for Multiple Graph Matching,” Machine Vision and Applications, vol. 27, no. 7, pp. 1021–1034, 2016.

24. Yongho Shin and **Kuk-Jin Yoon**, “PatchMatch Belief Propagation Meets Depth Upsampling for High-resolution Depth Maps,” Electronics Letters, vol. 52, no. 17, pp. 1445–1447, 2016.

23. Ju Hong Yoon, Ming-Hsuan Yang, and **Kuk-Jin Yoon**, “Interacting Multiview Trackers,” IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), vol. 38, no. 5, pp. 903–917, 2016.

22. Yongho Shin and **Kuk-Jin Yoon**, “Robust Spatiotemporal Stereo against Image Motion and Temporal Disparity Variation,” Electronics Letters, vol. 52, no. 7, pp. 515–517, 2016.

21. Seung Hwan Bae and **Kuk-Jin Yoon**, “Polyp Detection via Imbalanced Learning and Discriminative Feature Learning,” IEEE Transactions on Medical Imaging (TMI), vol. 34, no. 11, pp. 2379–2393, 2015.

20. Jungho Kim, **Kuk-Jin Yoon**, and In So Kweon, “Bayesian Filtering for Keyframe-based Visual SLAM", International Journal of Robotics Research (IJRR), vol. 34, no. 4-5, pp. 517–531, 2015.

19. Jonghee Park and **Kuk-Jin Yoon**, “Real-time Line Matching from Stereo Images using a Nonparametric Transform of Spatial Relations and Texture Information," SPIE Optical Engineering, vol. 54, no. 2, pp. 023106(1–11), 2015.

18. Min-Gyu Park, Jonghee Park, Yongho Shin, Eul-Gyoon Lim, and **Kuk-Jin Yoon**, “Stereo Vision with Image-guided Structured-light Pattern Matching,” IET Electronics Letters, vol. 51, no. 3, pp. 238–239, 2015.

17. Jong-Hee Park, Ju Hong Yoon, Min-Gyu Park, and **Kuk-Jin Yoon**, “Dynamic Point Clustering with Line Constraints for Moving Object Detection in DAS,” IEEE Signal Processing Letters (SPL), vol. 21, no. 10, pp.1255–1259, 2014.

16. Minkoo Kang and **Kuk-Jin Yoon**, “Depth-Discrepancy-Compensated Inter-Prediction with Adaptive Segment Management for Multiview Depth Video Coding,” IEEE Transactions on Multimedia (TMM), vol. 16, no. 6, pp. 1563–1573, 2014.

15. Seung Hwan Bae and **Kuk-Jin Yoon**, “Robust Online Multi-Object Tracking with Data Association and Track Management,” IEEE Transactions on Image Processing (TIP), vol. 23, no. 7, pp. 2820–2833, 2014.

14. Min-Koo Kang, Daeyoung Kim, and **Kuk-Jin Yoon**, “Adaptive Support of Spatial-Temporal Neighboring Samples for Depth Map Sequence Up-sampling,” IEEE Signal Processing Letters (SPL), vol. 21, no. 2, pp.150–154, 2014.

13. Ju Hong Yoon, Du Yong Kim, and **Kuk-Jin Yoon**, “Gaussian Mixture Importance Sampling Function for Unscented SMC-PHD Filter,” Signal Processing, vol. 93, no. 9, pp. 2664–2670, 2013.

12. Jae-changean Jeong, Ho-chul Shin, Jiho Chang, Eul-gyun Lim, Seungmin Choi, **Kuk-Jin Yoon**, and Jae-il Cho, “High-quality Stereo Depth Map Generation Using Infrared Pattern Projection,” ETRI Journal, vol. 35, no. 6, pp. 1011–1019, 2013.

11. Seung Hwan Bae, Du Yong Kim, Ju Hong Yoon, Vladimir Shin, and **Kuk-Jin Yoon**, “Automated Multi-target Tracking with Kinematic and Non-kinematic Information,” IET Radar, Sonar and Navigation, vol. 6, no. 4, pp. 272–281, 2012.

10. **Kuk-Jin Yoon**, "Stereo Matching based on Non-linear Diffusion with Disparity-Dependent Support-Weights", IET Computer Vision, vol. 6, no. 4, pp. 306–313, 2012.

9. Ju Hong Yoon, Du Yong Kim, and **Kuk-Jin Yoon**, "Efficient Importance Sampling Function Design for Sequential Monte Carlo PHD Filter", Signal Processing, vol. 92, no. 9, pp. 2315–2321, 2012.

8. Min-Gyu Park and **Kuk-Jin Yoon**, "Optimal Key-frame Selection for Video-based Structure-from-motion", Electronics Letters (EL), vol. 47, no. 25, pp. 1367–1369, 2011.

7. **Kuk-Jin Yoon** and Sung-Kee Park, “Improving Stereo Matching with Symmetric Cost Functions", IEICE Electronics Express, vol. 8, no. 2, pp.57–63, 2011.

6. **Kuk-Jin Yoon**, Emmanuel Prados, and Peter Sturm, "Joint Estimation of Shape and Reflectance using Multiple Images with Known Illumination Conditions", International Journal of Computer Vision (IJCV), vol. 86, no. 2-3, pp. 192–210, 2010.
5. Ji-Ho Cho, **Kuk-Jin Yoon**, and K. H. Lee, "Alpha-matte-based Depth Map Enhancement for Hairy Objects," Electronics Letters, vol. 46, no. 3, pp. 211–213, 2010.
4. **Kuk-Jin Yoon** and In So Kweon, "Distinctive Similarity Measure for Stereo Matching Under Point Ambiguity," Computer Vision and Image Understanding (CVIU), vol. 112, no. 2, pp. 173–183, 2008.
3. Sungho Kim, **Kuk-Jin Yoon**, and In So Kweon, "Object Recognition Using a Generalized Robust Invariant Feature and Gestalt's Law of Proximity and Similarity", Pattern Recognition (PR), vol. 41, no. 2, pp. 726–741, 2008.
2. **Kuk-Jin Yoon** and In So Kweon, "Adaptive Support-Weight Approach for Correspondence Search," IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), vol. 28, no. 4, pp. 650–656, 2006.
1. **Kuk-Jin Yoon** and In So Kweon, "Voting-based Separation of Diffuse and Specular Pixels," Electronics Letters, vol. 40, no. 20, pp. 1260–1261, 2004.

Conference Proceedings

179. Hyemin Yang*, Wooseong Jeong*, Giwon Lee*, and **Kuk-Jin Yoon** (* Equal Contribution), "Triggering Generalist Reasoning via Predictive Uncertainty for Dual-System VLA," NeurIPS 2026.
178. Hyun-Kurl Jang, Jihun Kim, and **Kuk-Jin Yoon** (* Equal Contribution), "Preference-Guided Adaptation for Open-Vocabulary Semantic Segmentation via Prompt Disagreement," NeurIPS 2026.
177. Jaewoo Jeong*, Seonkyu Song*, Hyeonwoo Park, Jegyeong Cho, Yujin Bae, Giwon Lee, Daehee Park, and **Kuk-Jin Yoon** (* Equal Contribution), "SRA: Spatial Reasoning Adapter via Evolving Social Interaction Graphs for Trajectory Prediction," NeurIPS 2026.
176. Hoonhee Cho*, Jae-Young Kang*, Giwon Lee*, Hyemin Yang*, Heejun Park*, and **Kuk-Jin Yoon** (* Equal Contribution), "VIPS: Vehicle-Infrastructure Cooperative Planning Benchmark via Pseudo-Simulation," ECCV 2026. (oral)
175. Jiyong Boo, Byeongin Joung, Hyemin Yang, and **Kuk-Jin Yoon**, "HSDF-Lane: Height-Aligned Signed Distance Field with Semantic Lane Prior for 3D Lane Detection," ECCV 2026.
174. Youngho Kim*, Hoonhee Cho*, and **Kuk-Jin Yoon** (* Equal Contribution), "GoStop: Reinforcement Learning for Adaptive Temporal Aggregation in Event-Based Feature Tracking," ECCV 2026.
173. Heejun Park, Jaeseok Jeong, and **Kuk-Jin Yoon**, "RAF: Reliability-Aware Fusion of Camera, LiDAR, and 4D RADAR for Robust 3D Object Detection in Adverse Weather," ECCV 2026.
172. Daehyun We, Youngho Yoon, Jiyong Boo, and **Kuk-Jin Yoon**, "MotionEditGS: Editing Motion and Appearance of 4D Scenes from Monocular Video via Semantically Anchored Gaussians," ECCV 2026.
171. Hyun-Kurl Jang*, Jihun Kim*, Hyeokjun Kweon*, and **Kuk-Jin Yoon** (* Equal Contribution), "Distill Once, Adapt Life-Long: Exploring Dataset Distillation for Continual Test-Time Adaptation," ECCV 2026.
170. Taewon Seo*, Seonae Jeon*, Giwon Lee*, **Kuk-Jin Yoon**†, and Daehee Park† (* Equal Contribution, † C0-corresponding Authors), "Unified Prediction and Planning via Conflict-Aware Disjoint Parameter Training," ECCV 2026.
169. Yujin Bae*, Jaewoo Jeong*, Hyeonseong Kim*, and **Kuk-Jin Yoon**, "Ego-human Motion Prediction with 3D-aware LLM," ECCV 2026.
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167. Minjun Kang, Inkyu Shin, Taeyeop Lee, Myungchul Kim, In So Kweon, and **Kuk-Jin Yoon**, "GeoNVS: Geometry Grounded Video Diffusion for Novel View Synthesis," ECCV 2026. (oral)

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26. Peter Sturm, Amaël Delaunoy, Pau Gargallo, Emmanuel Prados, **Kuk-Jin Yoon**, “3D and Appearance Modeling from Images,” 14th Iberoamerican Congress on Pattern Recognition (Invited Talk), 2009.
25. **Kuk-Jin Yoon**, Yekeun Jeong, and In So Kweon, ”Support Aggregation via Non-linear Diffusion with Disparity-Dependent Support-Weights for Stereo Matching,” ACCV 2009.
24. **Kuk-Jin Yoon**, Emmanuel Prados, and Peter Sturm, “Generic Scene Recovery using Multiple Images”, SSVM 2009.
23. Jean-Charles Bazin, **Kuk-Jin Yoon**, In So Kweon, Cedric Demonceaux, and Pascal Vasseur, “Particle Filter Approach Adapted to Catadioptric Images for Target Tracking Application,” BMVC 2009.
22. Jungho Kim, **Kuk-Jin Yoon**, and In So Kweon, “Robust 3-D Visual SLAM in a Large-scale Environment,” ISRR 2009.
21. Jean-Charles Bazin, Dang-Quang Pham, Inso Kweon, and **Kuk-Jin Yoon**, “Automatic Method for Closed Eye Correction,” ICIP 2009.
20. **Kuk-Jin Yoon** and In So Kweon, “Stereo Matching with the Distinctive Similarity Measure,” ICCV 2007.
19. **Kuk-Jin Yoon**, Amaël Delaunoy, Pau Gargallo, and Peter Sturm, “Toward Global and Model based Multiview Stereo Methods for Shape and Reflectance Estimation,” PACV 2007 (ICCV).
18. **Kuk-Jin Yoon** and In So Kweon, “Stereo Matching with Symmetric Cost Functions,” CVPR 2006.
17. **Kuk-Jin Yoon** and In So Kweon, “Correspondence Search in the Presence of Specular Highlights using Specular-Free Two-Band Images,” ACCV 2006.
16. **Kuk-Jin Yoon**, Yoo Jin Choi, and In So Kweon, “Fast Separation of Reflection Components Using a Specularity-Invariant Image Representation,” ICIP 2006.

15. Sungho Kim, **Kuk-Jin Yoon**, and In So Kweon, "Background Robust Object Labeling by Voting of Weight-Aggregated Local Features," ICPR 2006.
14. Sungho Kim, **Kuk-Jin Yoon**, and In So Kweon, "Object Recognition Using a Generalized Robust Invariant Feature and Gestalt's Law of Proximity and Similarity," IEEE Workshop on Perceptual Organization in Computer Vision 2006.
13. **Kuk-Jin Yoon** and In So Kweon, "Locally Adaptive Support-Weight Approach for Visual Correspondence Search," CVPR 2005.
12. **Kuk-Jin Yoon**, Yoo Jin Choi, and In So Kweon, "Dichromatic-Based Color Constancy Using Dichromatic Slope and Dichromatic Line Space," ICIP 2005.
11. **Kuk-Jin Yoon**, Yoo-Jin Choi, and In So Kweon, "Illuminant Chromaticity Estimation Using Dichromatic Slope and Dichromatic Line Space," FCV 2005.
10. **Kuk-Jin Yoon** and In So Kweon, "Human Perception Based Color Image Quantization," ICPR 2004.
9. Ouk Choi, **Kuk-Jin Yoon**, and In So Kweon, "A Hierarchical Window Based Approach for Correspondence Problem in Vision," ICMIT 2003.
8. **Kuk-Jin Yoon** and In So Kweon, "Color Image Segmentation Considering of Human Sensitivity for Color Pattern Variations," SPIE Photonics Boston 2001.
7. **Kuk-Jin Yoon** and In So Kweon, "Artificial Landmark Tracking Based on the Color Histogram," IROS 2001.
6. **Kuk-Jin Yoon** and In So Kweon, "Landmark Design and Real-Time Landmark Tracking for Mobile Robot Localization," SPIE Photonics Boston 2001.
5. **Kuk-Jin Yoon**, Gi-Jeong Jang, Sung-Ho Kim, and In So Kweon, "Fast Landmark Tracking and Localization Algorithm for the Mobile Robot Self-Localization," IFAC 2001.
4. **Kuk-Jin Yoon**, In So Kweon, Chan-Ho Lee, Jong-Kyu Oh, and In-Taek Yeo, "Landmark Design and Real-Time Landmark Tracking Using Color Histogram for Mobile Robot Localization," ISR 2001.
3. Jong-Eun Ha, Jin-Young Yang, **Kuk-Jin Yoon**, and In So Kweon, "Self-Calibration Using the Linear Projective Reconstruction," ICRA 2000.
2. **Kuk-Jin Yoon** and In So Kweon, "Moving Object Segmentation Algorithm for Human-like Vision System," HWRS 2000.
1. **Kuk-Jin Yoon**, In So Kweon, Chang-Yeong Kim and Yang-Seok Seo, "Moving Object Segmentation Based on Human Visual Sensitivity," BMCV 2000.

Patents

International

6. (Active) Kee Chang Lee, Wonhee Lee, **Kuk Jin Yoon**, Yongho Shin, Yeong Won Kim, Jin Ho Song, "Stereo Matching Method and Apparatus", US10853960B2, 2019.
5. (Active) Ji Hyo Lee, **Kuk Jin Yoon**, and Woo Sup Han, "Object Recognition System using Left and Right Images and Method", US9984280 B2, 2018.
4. (Active) Ji Ho Chang, Jae Chan Jeong, Ho Chul Shin, Dae Hwan Hwang, Seung Min Choi, Eul Gyoon Lim, Jae IL Cho, and **Kuk Jin Yoon**, "Stereo Matching Method and Device for Performing the Method", US9967516 B2, 2018.
3. (Active) Young-Su Moon, **Kuk Jin Yoon**, and Jae-hyun Kim, "Method and Apparatus for Estimating Image Optical Flow", US9811918 B2, 2017.
2. (Active) Sung-Hyun Kim, **Kuk-Jin Yoon**, Seung-Hwan Bae, Yeong-Jun Jo, Tae-Kyung Kim, and Ji-Woon Jung, "Polyp Detection Apparatus and Method of Operating the Same", US9530205 B2, 2014.

1. (Active) **Kuk-Jin Yoon**, Makoto Kimura, Jin-kyung Lee, Yasuo Takane, Tae-kyung Kim, Hyun-seok Choi, Taek-seong Jeong, Kyoung-hwan Moon, and Takashima Masahiro, “Method for Improving 3-Dimensional Effect and Reducing Visual Fatigue and Apparatus Enabling the Same", US8760502 B2, 2014.

Domestic

12. (Active) **Kuk-Jin Yoon**, Yeon Kun Lee, Jae Seok Jeong, Jong Seob Yun, Won June Cho, “Method and Apparatus of Detecting Object using Deep Learning of Omni Directionsl Image, Method of Autonomous Driving of Vehicle and Drone using the Method (전방향 영상의 딥러닝을 이용한 물체 감지 방법 및 장치, 그리고 이를 이용한 차량 자율 주행 방법 및 드론 자율 주행 방법),” KR 1024425450000
11. (Active) Inkyu Park, Incheol Cho,**Kuk-Jin Yoon**, Minkyu Cho, “Variable Photo-illumination Method of Photoactivated Gas Sensor for Gas Identification (가변광조사를 이용한 광활성 가스센서의 가스종 식별방법),” KR 1024562910000
10. (Active) Young-su Moon, **Kuk-Jin Yoon**, Jaehyun Kim, “Method and Apparatus for Estimating Image Optical Flow (영상 광류를 추정하는 방법 및 이를 위한 장치),” KR 1022653150000
9. (Active) Jonghyun Choi, S. Mohammad Mostafavi I., **Kuk-Jin Yoon**, “A Method and Apparatus for Generating Super Resolve Intensity Image (고해상도 강도 이미지 생성 방법 및 장치,” KR 1023661870000
8. (Active) Ji Ho Chang, Jae Chan Jeong, Ho Chul Shin, Dae Hwan Hwang, Seung Min Choi, Eul Gyoon Lim, Jae Il Cho, **Kuk-Jin Yoon**, “Method of Stereo Matching and Apparatus for Performing the Method (스테레오 매칭 방법 및 이를 수행하는 장치,” KR 1021789780000
7. (Active) **Kuk-Jin Yoon**, Dae-Yong Cho, “An Apparatus, a Method, and a Computer-readable Storage Medium for Refining Object Proposal (객체 검색 후보 영역을 개선하기 위한 방법, 컴퓨터-판독가능 저장 매체 및 장치),” KR 1018744710000
6. (Active) **Kuk-Jin Yoon**, Min-Gyu Park, “A Method, an Apparatus and a Computer-readable Storage Medium of Leveraging Stereo Matching with Confidence Measures (신뢰척도에 따른 레버리지 스테레오 매칭 방법, 장치 및 컴퓨터-판독 가능 저장 매체),” KR 1017685330000
5. (Active) **Kuk-Jin Yoon**, Jeong Gyun Lee, “Estimation Method and Apparatus for Information Corresponding Camera Orientation by Using Image (이미지를 이용하는 카메라의 방향과 관련한 정보 추정장치 및 추정방법),” KR 1018471130000
4. (Active) Ju Hong Yoon, Young Bae Hwang, Byeong Ho Choi, **Kuk-Jin Yoon**, “Image based Hierarchical Multiple Object Tracking Method and System using a Relative Motion between Objects (객체 간의 상대 움직임 정보를 이용한 영상 기반 계층적 다중 객체 추적 방법 및 시스템),” KR 1017683720000
3. (Active) **Kuk-Jin Yoon**, Sua Kim, “Method for 3D Object Detection and Pose Estimation (3차원 객체 검출 및 자세추정 방법),” KR 1018197300000
2. (Active) **Kuk-Jin Yoon**, Min-Gyu Park, “Confidence based Recursive Filtering Method for Depth Map Refinement (신뢰 기반 재귀적 깊이 영상 필터링 방법),” KR 1017633760000
1. (Active) Jihyyo Lee, **Kuk-Jin Yoon**, Woosup Han “Object Recognition System and Method the Same (물체 인식 시스템 및 그 물체 인식 방법),” KR 1017157820000

Research Projects (to be updated)

***PI** indicates Principal Investigator.*

110. **(PI)** Long-Horizon Agentic AI Research: National Research Foundation of Korea (NRF), 2B KRW per year, 07/2026 – 12/2031
109. Development of Outpainting AI and Spatial Expansion Technology for Utilizing Archive Video Contents: Institute for Information & Communications Technology Planning & Evaluation (IITP), 200M KRW, 04/2026 – 12/2026

108. Development of High-precision Hand-crafting and Service Technologies based on Physical AI: Institute for Information & Communications Technology Planning & Evaluation (IITP), 140M KRW, 04/2026 – 12/2026
107. **(PI)** Multi-modal Sensor based Self-improving Autonomous Driving Research: National Research Foundation of Korea (NRF), 332M KRW per year, 03/2026 – 03/2029
106. **(PI)** Development of a C-arm X-ray Video Engine: DK Medical Systems, 220M KRW, 07/2025 – 06/2026
105. **(PI)** Large-Scale Language Model Innovation Research Group: KAIST, 50M KRW, 07/2025 – 12/2025
104. Bio-Embodied Physical AI Research Group: DGIST, 385M KRW, 07/2025 – 12/2025
103. **(PI)** AI Robot-Based Digital Twin Care Platform for Continuous Integrated Care of the Elderly: Korea Health Industry Development Institute (KHIDI), 75M KRW, 05/2025 – 02/2026
102. **(PI)** Development of a Generative AI-Based 16-bit HDR Solution: Samsung Electronics, 66M KRW, 05/2025 – 04/2026
101. **(PI)** Automated Particle Analysis Technology Using Cathode SEM Images: LG Energy Solution, 99M KRW, 03/2025 – 10/2025
100. **(PI)** Research on Multimodal Camera-Based Computer Vision Technology for Robust Autonomous Driving: National Research Foundation of Korea (NRF), 206M KRW, 03/2025 – 02/2026
99. **(PI)** AI Research Hub Project: Institute for Information & Communications Technology Planning & Evaluation (IITP), 110M KRW, 01/2025 – 12/2025
98. **(PI)** Military-Specialized AI Curriculum Development and Operation (AI Leadership, Policy, Project Course): IITP, 31M KRW, 01/2025 – 12/2025
97. Development of Autonomous Driving Connectivity Technology Based on Sensor Infrastructure Cooperation: LG Electronics, 138M KRW, 01/2025 – 12/2025
96. **(PI)** Development of a Humanoid Robot Pilot Based on Natural Language Processing Knowledge Base: Agency for Defense Development (ADD), 102M KRW, 01/2025 – 12/2025
95. **(PI)** Graph-Based Approach to 3D Feature Point Tracking Algorithms of Event Cameras: KAIST, 2M KRW, 01/2025 – 06/2025
94. **(PI)** AI Robot-Based Digital Twin Care Platform for Continuous Integrated Care of the Elderly: KHIDI, 45M KRW, 11/2024 – 04/2025
93. **(PI)** Development of Data Augmentation and Sensor Fusion Technologies for Robust Autonomous Driving: 42dot, 500M KRW, 12/2024 – 02/2026
92. **(PI)** Samsung Electronics DS Division Strategic Industry-Academia Project (5th Year): Samsung Electronics, 57M KRW, 09/2024 – 09/2027
91. **(PI)** AI Research Hub Project: IITP, 55M KRW, 07/2024 – 12/2024
90. **(PI)** Object Prediction Model Using Surrounding Environment Perception and Map Information: Hyundai NGV, 88M KRW, 06/2024 – 05/2026
89. **(PI)** Research on Multimodal Camera-Based Computer Vision Technology for Robust Autonomous Driving: NRF, 186M KRW, 03/2024 – 02/2025
88. **(PI)** Research on Multimodal Camera-Based Computer Vision Technology for Robust Autonomous Driving: NRF, 18M KRW, 03/2024 – 02/2025
87. Development of Autonomous Driving Connectivity Technology Based on Sensor Infrastructure Cooperation: LG Electronics, 200M KRW, 01/2024 – 12/2024
86. Development of AI Sub-Modules for Intelligent Baggage X-ray Interpretation Training System: NRF, 89M KRW, 01/2024 – 06/2024
85. **(PI)** Samsung Electronics DS Division Strategic Industry-Academia Project (4th Year): Samsung Electronics, 57M KRW, 09/2023 – 08/2026

84. **(PI)** Development of Core Technologies for Autonomous Driving in Unstructured Off-Road Environments: Hanwha Aerospace, 165M KRW, 07/2023 – 06/2026
83. **(PI)** Research on Multimodal Camera-Based Computer Vision Technology for Robust Autonomous Driving: NRF, 206M KRW, 03/2023 – 02/2024
82. **(PI)** Research on Multimodal Camera-Based Computer Vision Technology for Robust Autonomous Driving: NRF, 2M KRW, 03/2023 – 02/2024
81. Development of a 2kg-Class Small EOTS for Drones: EOS System Co., Ltd., 60M KRW, 01/2023 – 11/2023
80. Development of a 2kg-Class Small EOTS for Drones: EOS System Co., Ltd., 2M KRW, 01/2023 – 11/2023
79. **(PI)** Development of a Humanoid Robot Pilot Based on Natural Language Processing Knowledge Base: ADD, 38M KRW, 01/2023 – 11/2023
78. Development of Real-Time Hologram Acquisition and Preprocessing Technology for 5G Services Based on Deep Learning: KETI, 100M KRW, 01/2023 – 12/2023
77. (DeepView – Main/Part 1) Development of a High-Performance Visual Discovery Platform: ETRI, 109M KRW, 01/2023 – 02/2024
76. **(PI)** Development of Artificial Intelligence Technology for Continuous Self-Improvement: IITP, 108M KRW, 01/2023 – 12/2023
75. Development of AI Sub-Modules for Intelligent Baggage X-ray Interpretation Training System: NRF, 46M KRW, 03/2023 – 12/2023
74. **(PI)** Military-Specialized AI Curriculum Development and Operation: IITP, 31M KRW, 06/2024 – 12/2024
73. **(PI)** Domain Adaptation for AI-based Detection and Monitoring: LIG Nex1, 100M KRW, 10/2022 – 09/2023
72. **(PI)** Low-level Image Fusion for Autonomous Driving: Hyundai Motors, 85M KRW, 11/2022 – 11/2023
71. **(PI)** Research on Multi-sensor Fusion for Autonomous Driving: NRF, MSIT, 850M KRW (4 years), 03/2022 – 02/2026
70. Development of Humanoid Robot Pilot based on NLP and Knowledge Base: ADD, 200M KRW per year, 01/2022 – 11/2026
69. AI Research for Intelligent X-ray Luggage Scanning Systems: NRF, 150M KRW per year, 07/2021 – 06/2024
68. **(PI)** Development of an Iron Plate Abrasion Rate Recognition System: Samsung Heavy industry, 60M KRW, 05/2021 – 01/2022
67. Development of Compact EOTS for Drones: EO Systems, 60M KRW per year, 01/2021 – 12/2023
66. **(PI)** Intelligent Focus Adjustment for Dual Pixel Cameras: Samsung Advanced Institute of Technology, 57.4M KRW per year, 09/2020 – 09/2023
65. **(PI)** Deep-learning-based 5G Real-time Hologram Generation and Processing: IITP, 120M KRW (per year), 2020 – 2023
64. **(PI)** Deep-learning-based Hand Pose Estimation using Low-resolution Images: KETI, 60M KRW, 05/2020 – 11/2020
63. **(PI)** Meta-fusion of Deep Neural Networks: ETRI, 90.9M KRW, 04/2020 – 11/2020
62. Research on Self-improving AI: NRF, 200M KRW (per year), 04/2020 – 12/2023
61. Deep View – Research on Vision- and Learning-based Scene Understanding and Event Forecasting: MSIP, 80M KRW (per year), 2019 – 2023.
60. Research on Autonomous Multi-agent CPS: ADD, 60M KRW per year, 12/2019 – 12/2024
59. Research on Mapping and Perception: NaverLabs, 85M KRW (per year), 2019 – 2022
58. Development of Quadruped Robot for Surveillance, Reconnaissance, and Search Missions: ADD, 140M KRW (per year), 12/2019 – 11/2024

57. **(PI)** Automatic Color Texture Generation for 3D Point Cloud Data: KETI, 80M KRW, 06/2019 – 11/2019
56. **(PI)** Vision-based Abnormal Event Detection: Hyundai Heavy Industries, 50M KRW, 04/2019 – 12/2019
55. **(PI)** AAVM Pedestrian Detection: Hyundai Construction Equipment, 60M KRW, 02/2019 – 12/2019
54. **(PI)** Computer Vision Algorithms based on 360° Cameras and Event Cameras: Naver Labs, 85M KRW, 01/2019 – 01/2020
53. **(PI)** Accurate Stereo Matching Algorithm for Indoor Robot Navigation: Samsung Research, 96M KRW, 05/2018 – 04/2019
52. Research on Multi-modal Hand Control: MOTIE, 140M KRW (per year), 04/2018 – 12/2019
51. **(PI)** Research on Event Camera-based Computer Vision Algorithms for Visual Intelligence: NRF, MSIT, 850M KRW (4 years), 03/2018 – 02/2022
50. Fundamental Study of Vision Algorithms for Comprehensive and Thorough Understanding of Videos: KRF, 235M KRW, 08/2017 – 04/2019
49. 4D Reconstruction of Non-rigid Dynamic Objects for Realistic Services: Giga KOREA Foundation, 04/2017 – 12/2020
48. 360° Stereo Camera-based Dynamic Scene Understanding for Autonomous Driving: Samsung Future Technology Foundation, 500M KRW, 09/2016 – 08/2019
47. **(PI)** AR/VR Platform Development for ADAS Research: GIST, 200M KRW, 05/2017 – 12/2017
46. **(PI)** Stereo-based High Speed and High Accurate Depth Sensing for AR HUD: Samsung Electronics, 90M KRW, 07/2016 – 06/2017
45. **(PI)** Visual Attention Estimation for VR: Samsung Electronics, 290M KRW, 05/2016 – 12/2017
44. Depth Sensing and Depth-based Road Monitoring: MOTIE, 110M KRW (per year on average), 03/2016 – 12/2017
43. **(PI)** Real-time Multi-Object Tracking: Hyundai Motors, 72.25M KRW, 12/2015 – 11/2016
42. **(PI)** Road Surface Inspection using Depth Images: Hyundai Mobis, 80M KRW, 09/2015 – 11/2016
41. **(PI)** Development of Local Stereo Matching Logic: KETI, 50M KRW, 09/2015 – 05/2016
40. **(PI)** Illumination and Reflection Estimation based on 3D Shape Analysis: ETRI, 50M KRW, 06/2015 – 01/2016
39. **(PI)** Stereo-vision-based 3D Dynamic Environment Analysis for Autonomous Driving: NRF, MSIP, 279M KRW (per year), 05/2015 – 04/2018
38. Online Monitoring and Extracting Features of Emotional Audience Responses: MCST, 50M KRW (per year), 04/2015 – 03/2018
37. Deep View – Research on Vision- and Learning-based Scene Understanding and Event Forecasting: MSIP, 45M KRW (per year), 03/2014 – 02/2018.
36. Real-time 3D Scene Modeling with Active Vision Sensors: MSIP, 50M KRW (per year), 09/2012 – 08/2015
35. **(PI)** Structure-from-motion for Mobile Devices: LG Electronics, 30M KRW, 09/2014 – 12/2014
34. **(PI)** Dynamic Scene Understanding using Stereo Cameras: Hyundai Mobis, 72M KRW, 05/2014 – 12/2014
33. **(PI)** High-Speed Optical Flow Estimation: Samsung Electronics, 70M KRW, 04/2014 – 01/2015
32. **(PI)** High Accuracy Stereo Vision with Pattern Projection: Samsung Electronics, 90M KRW, 11/2013 – 09/2014
31. **(PI)** Dynamic Objects Detection and Path Prediction using Stereo Cameras: LG Electronics, 60M KRW, 04/2013 – 12/2013
30. **(PI)** Automatic Polyp Detection in Endoscopic Images: Samsung Electronics, 85M KRW, 03/2013 – 12/2013
29. Interactive Performance based on Audience Reaction: MCST, 55M KRW, 08/2012 – 03/2013
28. **(PI)** Stereo with 2x2 Camera Array: LG Electronics, 50M KRW, 08/2012 – 07/2013

27. **(PI)** Stereo Matching Robust to Illumination Changes: ETRI, 50M KRW (per year), 05/2012 – 01/2015
26. **(PI)** Multi-baseline Stereo based SLAM for Dynamic Environments: MEST, 48M KRW (per year), 05/2012 – 04/2015
25. **(PI)** High-quality Disparity Map Estimation with Motion: Samsung Electronics, 90M KRW, 03/2012 – 02/2013
24. **(PI)** Endoscopic Image Processing: Samsung Electronics, 85M KRW, 04/2012 – 12/2012
23. **(PI)** Fusion of Active Laser Sensor and Camera: KIST, 30M KRW, 08/2011 – 08/2012
22. **(PI)** Sensor-fusion-based User Motion Capture: NHN and NIPA, 60M KRW, 11/2011 – 06/2012
21. **(PI)** Research on 3D Scene Reconstruction and Scene Flow Estimation: MEST, 50M KRW (per year), 05/2009 – 04/2012
20. Realistic Broadcasting Research Center (ITRC): NIPA, 20M KRW (per year), 01/2009 – 12/2011
19. **(PI)** High-resolution Depth Map Estimation using Semi-active Stereo: Samsung Electronics, 52M KRW, 04/2011 – 11/2011
18. **(PI)** Research on Terrain Matching Methods for Terrain-aided Navigation (TAN): LIG Nex1/ADD, 50M KRW, 06/2010 – 05/2011
17. **(PI)** System Development for Illumination Source Estimation: Viewrun and ETRI, 40M KRW, 09/2010 – 06/2010
16. **(PI)** Development of Automatic Inter-Camera Distance Adjustment Methods: Samsung Electronics, 60M KRW, 03/2010 – 12/2010
15. **(PI)** Research on Texture Synthesis and Specular Reflection Removal: ETRI, 60M KRW, 06/2009 – 01/2010
14. **(PI)** Object Contour Extraction for Robot Grasping: KIST, 25M KRW (per year), 01/2009 – 12/2010
13. Development of Experience Tour Technology based on Mobile Mixed Reality: KIST, 20M KRW, 03/2009 – 02/2010
12. **(PI)** Multi-view Image Stitching: Samsung Advanced Institute of Technology, 30M KRW, 07/2009 – 07/2010
11. **(PI)** Object Recognition with Stereo Cameras: Samsung Electronics, 150M KRW, 02/2009 – 12/2009
10. Flamenco Project: ANR (France), 2007 – 2008
9. Robust Robot Vision Research: MOST National Research Laboratory, 2003 – 2006
8. Vision-based Environments Recognition for Network-based Humanoids: KIST, 2004 – 2006
7. Development of the Real-Time 3D Image Sensor: Samkyung Hitech, 2001 – 2003
6. Vision Guidance System based on Human Binocular Vision Model: BSRC, 2001 – 2003
5. Development of Entertainment Robots: HWRS-ERC, 1999 – 2003
4. Imaging System for 3D Display: KIST, 2002
3. Vision for Mobile Robot: Samsung Electronics, 2002
2. Image-based Guidance System for AGV: Hyundai Heavy Industry, 2000
1. Image/Video Indexing: SAIT, 2000

Teaching (in English)

- Fall 2025: Autonomous Mobile Systems Programming
- Spring 2025: Introduction to Visual Intelligence
- Fall 2024: Autonomous Mobile Systems Programming
- Spring 2023: Introduction to Visual Intelligence

- Fall 2022: Autonomous Mobile Systems Programming
- Spring 2022: Introduction to Visual Intelligence
- Fall 2021: Special Topics in Mechanical Engineering - Programming for Autonomous Mobile Systems
- Spring 2021: Introduction to Visual Intelligence
- Spring 2021: Capstone Design I
- Fall 2020: Special Topics in Mechanical Engineering - Programming for Autonomous Mobile Systems, MyME, Capstone Design II
- Spring 2020: Special Topics in Mechanical Engineering - Visual Intelligence, MyME
- Fall 2019: Programming for Autonomous Mobile Systems, MyME, Capstone Design II
- Spring 2019: Visual Intelligence, MyME, Capstone Design I
- Fall 2018: Random Data, Programming for Autonomous Mobile Systems
- Spring 2018: Visual Intelligence
- Fall 2017: Computer Vision
- Spring 2017: Signals and Systems
- Fall 2016: Computer Vision
- Spring 2016: Digital Image Processing
- Fall 2015: Computer Vision
- Spring 2015: Signals and Systems
- Fall 2014: Computer Vision
- Spring 2013: Signals and Systems
- Fall 2012: High-level Image Understanding & Processing – Computer Vision
- Spring 2012: Signals and Systems
- Fall 2011: Digital Image Processing
- Spring 2011: High-level Image Understanding & Processing – Computer Vision
- Fall 2010: Digital Image Processing
- Spring 2010: High-level Image Understanding & Processing – Computer Vision
- Fall 2009: Digital Image Processing
- Spring 2009: High-level Image Understanding & Processing
- Fall 2008: Digital Image Processing

Invited Talks (to be updated)

International

10. **Plenary Talk**: "Visual Sensing and Perception for Autonomous Driving," IFAC Conference on Engin and Powertrain Control, Simulation and Modeling (E-COSM), Oct. 31, 2024
9. (Invited Talk) "Event Camera-based Computer Vision," DeepView Workshop at AVSS 2022, Online, Nov. 2022
8. (Department Seminar) "Computer Vision with Omnidirectional and Event Cameras," AI Thrust Seminar at HKUST, Online, April 2022
7. (Invited Talk) "Sensing and Perception with 360° and Event Cameras for Autonomous Driving," International Symposium on Future Mobility (ISFM), 2019
6. (Invited Talk) "Applying Deep Learning to 360° and Event Cameras," DGIST Global Innovation Festival, Korea, 2019

5. (Invited Talk) “Generating Content-aware Perspective Videos from 360° Videos for Comfortable 360° Video Watching,” 24th International Workshop on Frontiers of Computer Vision, Japan, 2018
4. (Invited Talk) “Generating Content-aware Perspective Videos from 360° Videos for Comfortable 360° Video Watching,” DGIST Global Innovation Festival, Korea, 2017
3. (Invited Talk) “Robust Stereo Matching with Temporal Aggregation and Matching Confidence,” International Conference on Internet of Vehicles, Nadi, Fiji, 2016
2. (Invited Talk) “How Much Further Can We Go in Two-frame Stereo?”, Symposium on High Precision Stereo Vision, SIAM IS 2014, Hong Kong, 2014
1. (Invited Paper) Peter Sturm, Amaël Delaunoy, Pau Gargallo, Emmanuel Prados, **Kuk-Jin Yoon**, “3D and Appearance Modeling from Images,” 14th Iberoamerican Congress on Pattern Recognition, 2009.

Domestic

96. “대학 현장에서의 AI 활용 방향과 미래 인재 양성 방안(Directions for AI Utilization in Higher Education and Future Talent Development)," 전국대학교기획처장협의회(Council of University Planning Directors), Dec. 18, 2025
95. “국가 인공지능 계획과 연계한 KAIST의 인공지능 전략과 대전 인공지능 산업 발전 방향(KAIST’s AI Strategy and the Future of Daejeon’s AI Industry in Alignment with the National AI Plan)," 대전인공지능포럼(Daejeon Artificial Intelligence Forum), Dec. 16, 2025
94. “From Perception to Action: Visual Intelligence to Physical AI," PCN, Dec. 9, 2025
93. “How to do Good Research: Effective Strategies for Paper Preparation," UNIST, Sep. 19, 2025
92. “From Perception to Action Visual Intelligence to Physical AI," Osstem Impant, Sep. 5, 2025
91. “How to do Good Research: Effective Strategies for Paper Preparation," Hanyang Univ., Aug. 26, 2025
90. “How to do Good Research: Effective Strategies for Paper Preparation," DGIST, Jul. 21, 2025
89. “Event-Based Vision: A High-Speed Alternative to Frame Cameras," DGIST ME Department Seminar, Apr. 8, 2025
88. “Event-Based Vision: A High-Speed Alternative to Frame Cameras," KAIST Kim Jaechul Graduate School of AI, Department Seminar, Apr. 1, 2025
87. “Visual Sensing and Perception for Autonomous Driving," The 20th Korea Robotics Sociery Annual Conference (KRoC 2025), Invited Talk, Feb. 13, 2025
86. “Visual Sensing and Perception for Autonomous Driving," Winter School of Image Understanding Research Group of IEIE Korea, Invited Talk, Jan. 14, 2025
85. “How to do Good Research: Effective Strategies for Paper Preparation," POSTECH Department Seminar, Nov. 05, 2024
84. “How to do Good Research: Effective Strategies for Paper Preparation," SNU IPAI Seminar, Oct. 11, 2024
83. “Multi-modal Foundation Model for Autonomous Driving,” The Korean Society of Automotive Engineers Workshop for the New Paradigm for Autonomous Driving, Oct. 08, 2024
82. “Computer Vision for Autonomous Mobility," SNU ME Department Seminar, March 2023
81. Workshop on Vision & Graphics AI and Acceleration for Self-driving Cars: Seeing for Moving – Computer Vision with 360-degree Cameras and Event Cameras for Autonomous Driving, 02/2022
80. GIST EECS Colloquium: Seeing for Moving: Computer Vision for Smart Mobility, 12/2021
79. Republic of Korea Air Force Headquarters: Seeing for Moving – Introduction to Artificial Intelligence, 10/2021
78. 수중수상로봇연구회 기조강연: View More Widely and Clearly – Scene Perception with 360-degree Cameras and Event Cameras, 05/2021

77. 2nd Operations Command: Artificial Visual Intelligence and Its Applications, 01/2021
76. (Plenary Talk) Korean Conference on Computer Vision (KCCV) 2020: Computer Vision and Machine Learning for Autonomous Driving, August 2020.
75. 2020 Software Convergence Symposium(SWCS2020: Scene Understanding using 360° and Event Cameras, August 2020.
74. LIG Nex1: Computer Vision and Machine Learning based 3D Dynamic Scene Understanding, Januray 2020.
73. ICROS-KROS 대전충청: ADAS for Autonomous Driving based on Computer Vision and Machine Learning, December 2019.
72. 한국자동차공학회: 강인한 자율주행을 위한 360도 이벤트 카메라 응용 연구, October 2019.
71. ADD: 360도 카메라 및 이벤트 카메라를 활용한 딥러닝 기반의 환경 인식 연구, September 2019.
70. KAIST 문술미래전략대학원: 미래도시: 자율주행자동차, June 2019.
69. 삼성전기: Research on event camera-based computer vision algorithms for visual intelligence, June 2019
68. Postech: Applying Deep Learning to 360° and Event Cameras, April 2019.
67. 연세대학교: DNNs for 360° and Event Cameras, January 2019.
66. KCCV 2018 (invited): Joint Layout Estimation and Global Multi-view Registration for Indoor Reconstruction, July 2018.
65. ETRI: Multi-camera Network Topology Estimation and Person Re-ID, May 2018.
64. GIST: 컴퓨터 비전 및 기계학습 기반 자율주행을 위한 요소 기술, April 2018.
63. KIST: 360° Videos and ADAS, April, 2018.
62. 네이버랩스(Naver Labs): Survey on Lane-Level Localization, July 2017.
61. Vivozon: Computer Vision-based Scene Understanding, September 2017.
60. 개방형컴퓨터통신연구회(OSIA): 자율 주행을 위한 컴퓨터 비전 및 머신 러닝 기반 주행 환경 인식 기술, June 2017.
59. KCCV 2017 (invited): Multi-attributed Graph Matching with Multi-layer Random Walks, June 2017.
58. 네이버랩스(Naver Labs): 영상에서의 Appearance 및 움직임 정보 모델링을 통한 다중 객체 추적, March 2017.
57. 경희대학교: 자율주행 자동차를 위한 비전 기반 ADAS 연구, November 2016.
56. 자동차 융합 얼라이언스 기술 발전 세미나: 자율주행을 위한 컴퓨터 비전 기반 동적 주행 환경 인식 기술, October 2016.
55. KCCV 2016 (invited): Tracking and Identifying Multiple Objects across Multiple Cameras, July 2016.
54. 대한전자공학회 영상처리연구회 워크샵: Dynamic 환경에서의 자율 주행체를 위한 비전 기반 응용 기술, July 2016.
53. 스마트카 센서/부품 테크포럼 세미나 2016: 컴퓨터 비전 기반 동적 주행 환경 인지 기술, June 2016.
52. UMV 자율주행기술 전문가 세미나: 자율 이동체를 위한 영상 기반 상황 센싱 및 인지 기술, June 2016.
51. 한국미래기술교육연구원 세미나: 스마트카의 자율주행을 위한 스테레오 영상 기반 동적 상황 인지 기술 연구, April 2016.
50. 호남 ETRI, March 2016.
49. 대한전자공학회 컴퓨터비전 튜토리얼, February 2016.
48. ETRI, Daejeon, December 2015.
47. Hyundai Mobis 기술포럼 전문가 세미나: 차량용 카메라 보정을 위한 자세 추정 기법 및 주변 방애물 검출을 위한 3차원 복원 방법, November 2015.

46. ETRI: Multiple Object Tracking, Daejeon, September 2015.
45. KAIST NOVIC Seminar: 스마트카의 자율주행을 위한 컴퓨터 비전 기반 동적 상황 인지 기술, September 2015.
44. KCCV 2015 (invited): Leveraging Stereo Matching with Learning-based Confidence Measures, August 2015.
43. POSTECH: Multi-object Tracking Tutorial, August 2015.
42. KAIST: 스마트카의 자율주행을 위한 스테레오 영상 기반 ADAS 기술, August 2015.
41. 정보과학회 CVPR 워크샵: 스마트카의 자율 주행을 위한 동적 상황 인지 기술, July 2015.
40. IPIU 2015 초청논문: 카메라 움직임에 강건한 영상기반 다중객체 추적 방법, February 2015.
39. 한국에너지기술연구원: Fourier and Wavelet Transform, January 2015.
38. SK Telecom: Robust Online Object Tracking, December 2014.
37. SNU: Multi-object Tracking Tutorial, November 2014.
36. 자동차 공학회 전기전자ITS 부문 워크샵: Vision-based Moving Obstacle Avoidance for Autonomous Vehicles, October 2014.
35. KCCV 2014 (invited): Online Robust Multi-target Tracking, August 2014.
34. 정보과학회 여름학교: Geometric Computer Vision, August 2014.
33. KETI: Recent Advances on Online Robust Multi-target Tracking, August 2014.
32. Hanyang Univ.: Stereo, April 2014.
31. POSTECH, Pohang, (Department Seminar) 2014
30. Yonsei University, Seoul, January, 2014
29. KIST, Seoul, September, 2013
28. KIST, Seoul, July 2013
27. KETI, Bundang, May 2013
26. POSTECH, Pohang , March 2013
25. Inha Univ., Incheon , January 2013
24. KICT, Goyang-si, December 2012
23. Dongseo Univ., Busan, November 2012
22. ETRI, Daejeon, May 2012: Semi-active Stereo Vision
21. Korean Society of Broadcast Engineers, Seoul, August 2011: Tutorial on Stereo Vision
20. Pentech, Seoul, July 2011: Computer Vision for Mobile Devices
19. GIST Science School, Gwangju, November 2010
18. Yeungnam Universityl, Kyungsan, November 2010
17. Agency for Defense Development(ADD), Daejeon, July 2010: Tracking Filters for Terrain-Aided Navigation
16. Samsung Electronics, Suwon, May 2010: Introduction to Stereo Vision and 3D Reconstruction
15. LIG NEX1, Suwon, April 2010: Terrain-Aided Navigation
14. Electronics and Telecommunications Research Institute (ETRI), Daejeon, Korea, April 2010.
13. Electronics and Telecommunications Research Institute (ETRI), Daejeon, Korea, July 2009.
12. Daegu Gyeongbok Institute of Science and Technology (DGIST), Daejeon, Korea, December 2008.
11. Korea Institute of Science and Technology (KIST), Seoul, Korea, December 2008.
10. Department of Electrical Engineering and Computer Science, KAIST, Daejeon, Korea, October 2008: Stereo Vision

9. Department of Information and Communications, GIST, Gwangju, Korea, February 2008: Multi-view Stereo under Image Ambiguity and Appearance Changes
8. Perception Team in INRIA Rhône-Alpes, Montbonnot, France, September 2006: Stereo Matching under Image Ambiguity and Appearance Changes
7. The 4th KAIST–Tsinghua Joint Workshop on Pattern Recognition, Daejeon, Korea, September 2005: Reflection Analysis using a Single Color image and Its Application to Stereo
6. The 1st International Joint Workshop of KAIST–RCV and U.Tokyo–Ikeuchi Lab. on Robust Vision Technology, Daejeon, Korea, April 2005: Robust Vision Techniques based on the Local-Level Analysis of Image Information
5. The 3th KAIST–Tsinghua Joint Workshop on Pattern Recognition, Beijing, China, December 2004: Locally Adaptive Support-Weight Approach for Visual Correspondence Search
4. Samsung Advanced Institute of Technology, December 2003: Stereo Vision
3. NRL(National Research Laboratory) Joint Workshop on Intelligent Robot Technology, Kyungju, Korea, October 2003: 3D Computation, Obstacle Detection/Avoidance, and Object Tracking using Stereo Vision for Intelligent Robots
2. The 6th Autumn Seminar of a Korean Society for the 3D Medical Image Research, September 2001: Tutorial on the 3D Modeling from Multiple Images
1. Advanced Science Institute 2001, Tokyo, Japan, July 2001: Computer Vision Applications

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